Likelihood-Preserving Compression for Coded SP-AFDM

Structural theory, an executed certified reference method, and an independent research verdict

Research continuation based on the supplied manuscript and receiver handoff

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Research verdict

Selected research line: preserving the decoding likelihood under a computation-limited representation, with explicit treatment of relative Doppler and the superimposed-pilot linear term. This replaces the narrower question of how accurately a receiver can approximate the physical channel matrix with a few integer atoms.

The executed reference method is **certified likelihood sparsification**: retain the exact matched linear statistic, sparsify the Hermitian interaction matrix, and bound the resulting score oscillation over the allowed codewords. This yields pointwise posterior, bit-LLR, and exact-MAP preservation statements. The basic matched-filter transformation, coefficient thresholding, and bounded-energy probability inequalities are not claimed as inventions.

Publication verdict: the research problem is defensible, but neither the old receiver nor the newly tested certified reference has earned a new Transactions-level superiority claim in this pass. The strongest positive compression results have a simpler classical explanation: matched-filter-domain detection, and, for common fractional offsets, ordinary derotation. The certificate is mathematically valid and useful for analysis, but the implemented exact-code policy does not demonstrate a computational advantage over the full small-system reference. A genuinely new scalable method remains unconfirmed. This is a scientific outcome, not an invitation to present an unsupported positive story.

Three results are worth carrying into a paper-development process. First, a dense fractional-Doppler AFDM operator can induce an exactly sparse likelihood interaction matrix; absolute fractional offset and likelihood complexity are not equivalent. Second, posterior-score comparison provides a pointwise guarantee even when both scores use the same observation-dependent channel estimate. This avoids misusing an exogenous generative KL bound. Third, a useful certification policy must price its own repeated inference and verification work. Saving graph entries alone does not establish a low-cost receiver.

The alternative directions were handled as follows. Integer expansion remains a valid representation and strong baseline. A generic communication-weighted projection is not selected: it is an oracle construction, is connected to established channel-shortening theory, and did not provide dependable coded gains in these controls. Adding diagonal posterior variances to the existing noncollision channel-estimation interface collapses to scalar ridge. Richer correlations remain potentially useful, but trustworthy acquisition is unresolved. The recent finite-tail receiver is not promoted: the supplied latest comparison does not establish superiority over ridge on the same expanded support. Continuous/off-grid estimation remains a required comparator when pursuing physical-parameter recovery, not an assumed necessity for communication.

Sources, audit, and evidence boundaries

What was read and reused

The supplied theoretical chapter, the handoff state and derivation documents, both September tail reports, and the physical and receiver interfaces were inspected. All **21 entries** in the supplied SHA-256 manifest match their bytes and sizes. The manifest describes a working-tree snapshot at repository HEAD `cdc4daac`, not a clean commit archive. Its header is dated September 17, while the included theoretical PDF is dated September 19. The package therefore preserves content hashes and records this chronological inconsistency rather than inferring a precise historical sequence from filenames.

The code was classified into three roles:

Role	Assets	Treatment
System/reference	DAFT conventions, G1 physical operator, coding construction	Independently checked; selected definitions reused without editing
Frozen benchmark	`direction_receiver`, feedback interfaces, checkpoint-bound initialization	Left unchanged; no claim of full replay
Historical exploration	`classical_joint_unrolled`, alternative baselines, narrative audits	Read as evidence and possible controls, not as current instructions

The full I1 environment is not present. Internal imports such as `supplement_pilots` and other modules are missing from this partial handoff; the checkpoint named by the frozen adapter is also absent. The machine-readable audit enumerates missing internal dependencies. No full I1 execution, version-matched production FER replay, or production cost comparison is claimed.

Two mature assets were reused. The standalone `physical_channel_matrix` function was extracted as an **unaltered AST function** from the G1 source and executed with NumPy. This bypasses unrelated missing imports without inventing their implementations. The original `e1_ldpc.py` was imported to construct and encode tiny Gallager codes. Both uses are recorded by source hash. The experiment then performs complete codeword enumeration, not the original finite-iteration production decoder.

What the latest project evidence changes

The September 16/17 handoff reports already include full G1 propagation. At 7 dB, the original, TAIL, and same-expanded-support ridge arms recorded 13, 9, and 11 failed frames out of 32. At 8 dB, the corresponding counts are 2, 0, and 1. The reported TAIL-minus-ridge paired interval includes zero, and the report's promotion decision is negative. A later sign-test sensitivity analysis further qualifies the original-versus-TAIL comparison. Those reports also record a frame on which better channel NMSE coincides with worse response error, LLR log loss, and decoding.

These are **source-reported results**, not numerical results rerun here: their original production arrays are not in the handoff. They rule out saying that full tails have never been explored. They do not prove that all tail-aware methods fail, or that ridge universally dominates.

The previous theoretical chapter remains useful. Its finite integer expansion, operator orthogonality, SP nuisance covariance, fixed-working-distribution channel-estimation identity, and exogenous model-

divergence qualifications are retained. The present report develops a different downstream object rather than treating the chapter's candidate receiver as mandatory.

New execution scope

The main reference was run at $N = 8$ and $N = 12$, with 90 fixtures per size and 256 observations per fixture. The two reused code constructions have 64 and 256 codewords respectively. There are 46,080 evaluated oracle-channel observation/configuration instances and 768 additional observation-driven channel-estimation diagnostic frames. These are **not** 46,080 independent fading realizations: messages/noise are deliberately reused across power, noise, and channel-regime controls. Each regime contains only three selected channel-phase seeds.

The numerical checks comprise 859 assertions per block size, 8,186 saved-array verification assertions, and 540 equivalence checks of a separately written sparse/incremental policy. Many are repeated identity checks, not independent scientific replications. The simulator and independent checks each finish in seconds in the recorded CPU environment. No large training or production Monte Carlo campaign was launched.

Problem formulation and exact physical model

Let b be an information message, $d(b) \in \mathbb{C}^N$ its coded QPSK vector, and p a deterministic known superimposed pilot. The physical observation is

$$x = p + d(b), \qquad y = H(\theta)x + w, \qquad w \sim \mathcal{CN}(0, \sigma^2 I). \tag{1}$$

Pilot and data vectors include their power scaling. In the new experiments, $\mathbb{E}|d_i|^2 = 1 - \rho$ and $|p_i|^2 = \rho$; $\mathbb{E}|w_i|^2 = \sigma^2$. The average total input energy is one per useful sample because the codebook has zero mean. No per-arm or post-truncation energy normalization is performed. A frame's realized pilot-data cross term need not vanish.

The operational target is message error risk subject to complete receiver cost:

$$\min_{\mathcal{A}} \Pr\{\widehat{b}_{\mathcal{A}}(y) \neq b\}, \qquad \mathbb{E}C(\mathcal{A}) \leq B. \tag{2}$$

The principal model assumes integer sample delays, fractional Dopplers retained without truncation, constant path parameters within the useful block, a sufficient chirp-periodic prefix (CPP), ideal synchronization, and proper sampled white noise. Fractional delays, pulse-shaping mismatch, time-varying parameters within the block, and amplifier nonlinearities are outside the proofs below.

Sign convention reconciled with the code

The handoff uses positive chirps and positive Doppler, with a negative-exponent unitary DFT:

$$F_{mn} = N^{-1/2} e^{-j2\pi mn/N}, \qquad A = \Lambda_{c_2}^{+} F \Lambda_{c_1}^{+}, \qquad \Lambda_c^{+} = \mathrm{diag}(e^{j2\pi c n^2}). \tag{3}$$

Define $[\Pi_l x]_n = x_{(n-l)_N}$, $\Delta_\nu^{+} = \mathrm{diag}(e^{j2\pi\nu n/N})$, and

$$[C_l]_{nn} = \begin{cases} e^{j2\pi c_1[N^2 - 2N(l-n)]}, & n < l, \\ 1, & n \geq l. \end{cases} \qquad U(l,\nu) = A C_l \Delta_\nu^{+} \Pi_l A^H. \tag{4}$$

Then $H = \sum_{r=1}^{P} h_r U(l_r, \nu_r)$. Every individual path operator is unitary. The prior manuscript's negative-chirp/negative-Doppler convention represents this same operator after the consistent substitution $(c_1, c_2, \nu) \mapsto (-c_1, -c_2, -\nu)$. Keeping all numerical parameters unchanged while switching signs is not equivalent.

The unmodified G1 FFT-kernel implementation agrees with direct sample/CPP matrix construction, including nontrivial prefix phases. An independent negative-convention reference with the proper sign mapping agrees with all saved physical channels to at most 3.83×10^{-15}. Naively reusing the same signed parameters gives a Frobenius discrepancy of at least 4.03 on the saved fixtures. This is a convention mismatch, not evidence that G1 is physically defective.

Integer representation remains exact when complete

For the positive Doppler convention,

$$a_q(\nu) = \frac{1}{N} \sum_{n=0}^{N-1} e^{j2\pi(\nu-q)n/N}, \qquad U(l, \nu) = \sum_{q=0}^{N-1} a_q(\nu) U(l, q). \tag{5}$$

Proposition 1. Equation (5) is exact; $\sum_q |a_q|^2 = 1$. Distinct integer operator coordinates are Frobenius-orthogonal:

$$\mathrm{tr}[U(l, q)^H U(l', q')] = N \mathbf{1}_{l=l'} \mathbf{1}_{q=q'}. \tag{6}$$

Proof. Equation (5) is finite Fourier inversion of the sampled modulation; Parseval gives the energy. Unitary conjugation preserves the trace product. Different delays leave a nonzero circular shift with zero diagonal. Equal delays cancel the prefix factors, leaving a complete integer-frequency exponential sum. This establishes (6).

Contributions sharing the same integer coordinate must be merged coherently. If $g_{lq} = \sum_{r:l_r=l} h_r a_q(\nu_r)$, the best Frobenius projection on support $\mathcal{S}$ has error $N \sum_{(l,q)\notin\mathcal{S}} |g_{lq}|^2$. A small physical path count therefore need not imply a small integer coefficient count. Conversely, exact representability does not make N^2 arbitrary coefficients identifiable from one N-sample observation.

Effective-shift collisions are a different issue. When $\alpha = 2Nc_1$ is integer, each integer operator is monomial, but distinct (l, q) can induce the same cyclic shift. Collisions can change graph degree and posterior-moment structure without destroying operator-space orthogonality. This distinction motivated the second block-size experiment.

SP uncertainty and the channel-estimation interface

For initial data covariance V_0, the exact nuisance second moment is

$$\mathrm{Cov}(Hd + w \mid H) = HV_0H^H + \sigma^2 I. \tag{7}$$

This does not imply Gaussian data interference. Static multipath can produce colored interference; a single fractional-Doppler unitary path with isotropic data produces white covariance. Truncated and retained data contributions are generally correlated through $H_R V_0 H_T^H$. These established distinctions from the supplied chapter remain in force.

For a frozen working distribution $q(d)$, let $\mu = \mathbb{E}_q d$, $V = \mathrm{Cov}_q(d)$, $H(g) = \sum_i g_i U_i$, and $\Phi_\mu = [U_i(p+\mu)]_i$. Direct expansion gives

$$\mathbb{E}_q \|y - H(g)(p+d)\|^2 = \|y - \Phi_\mu g\|^2 + g^H \Gamma g, \quad \Gamma_{ij} = \mathrm{tr}(U_i^H U_j V). \tag{8}$$

With a proximal penalty, the normal equation is

$$(\Phi_\mu^H \Phi_\mu + \Gamma + \lambda I)\widehat{g} = \Phi_\mu^H y + \lambda g_o. \tag{9}$$

Proposition 2. If V is diagonal and the active monomial operators have noncolliding effective shifts, then $\Gamma = \mathrm{tr}(V) I$.

Proof. For $[U_i x]_n = u_i[n] x_{n-s_i}$ with $|u_i[n]| = 1$,

$$\Gamma_{ij} = \sum_n u_i[n]^* u_j[n] V_{n-s_j, n-s_i}. \tag{10}$$

Different shifts select off-diagonal entries of diagonal V, which vanish. Equal indices contribute $\sum_n V_{nn}$. Shift collisions or genuine cross-symbol covariance invalidate the scalar reduction.

The new independent audit checks (8) by exact codeword averaging and verifies (10). In deliberately high-uncertainty frames, the exact posterior Γ has off-diagonal norms 0.2551 and 0.4055 at $N = 8, 12$, while the diagonal-covariance version collapses to scalar ridge to numerical precision. The high-entropy frames were selected solely to exercise the algebra, not to establish a receiver gain. Full posterior enumeration is not a low-cost source of production correlation statistics.

Equation (9) is an exact minimizer of the specified frozen quadratic. It is an exact EM M-step only when paired with the appropriate exact E-step and complete-data model; approximate decoder moments do not grant full marginal-likelihood or FER monotonicity. These are established inference ingredients [R2, R3, R4], not new contributions.

Change the compressed object: the decoding likelihood

Freeze an observed y and a channel operator $\bar{H}$. The operator may be true CSI or a plug-in estimate produced from that same observation. Define

$$R = \bar{H}^H \bar{H}, \qquad z = \bar{H}^H (y - \bar{H} p). \tag{11}$$

For codewords with positive prior $\pi(d)$, the plug-in log score, up to a codeword-independent term, is

$$s(d) = \frac{2\Re(d^H z) - d^H R d}{\sigma^2} + \log \pi(d). \tag{12}$$

Proposition 3. Equation (12) induces exactly the same finite-code posterior and MAP decision as $-\|y - \bar{H}(p+d)\|^2/\sigma^2 + \log \pi(d)$.

Proof. Expand the squared norm of $(y - \bar{H}p) - \bar{H}d$. The first term is independent of d; the cross term and quadratic term are exactly (12).

This is the classical matched-filter/Ungerboeck likelihood representation. With known H, the noise in z has covariance $\sigma^2 R$, not $\sigma^2 I$. Equation (12) is obtained by expansion of the original likelihood, not by pretending that the matched-filter output is an independent white-noise observation. This distinction avoids an incorrect statistical model.

The pairwise factor for coordinates i, j is proportional to $\exp[-2\Re(d_i^* R_{ij} d_j)/\sigma^2]$. A sparse Hermitian approximation R_B deletes selected interactions while preserving the exact linear vector z. This defines

$$s_B(d) = \frac{2\Re(d^H z) - d^H R_B d}{\sigma^2} + \log \pi(d). \tag{13}$$

No positive-definiteness assumption on R_B is required for this finite-code score to normalize: the codebook is finite and all scores are finite. Such an approximation should not be reinterpreted automatically as a valid continuous Gaussian channel covariance.

Constant-modulus equivalence and the SP caveat

Proposition 4. For $|d_i|^2 = P_d$ for every codeword and coordinate, replacing the diagonal of R by any real diagonal leaves the posterior unchanged, provided z remains fixed.

Proof. The diagonal change contributes the same $P_d \sum_i \Delta R_{ii}$ to every quadratic energy, which cancels during normalization.

This does not justify using the modified diagonal in Rp. The linear statistic in (11) must first be computed from the original model. Similarly, using $R_B p$ in place of Rp introduces an additional linear score error even if all data symbols have constant modulus. Nonconstant-modulus QAM does not have the arbitrary-diagonal invariance.

Thus two distinct objects should be preserved separately: the matched linear statistic and the codeword-dependent quadratic interactions. A single compressed channel G ties them together as $G^H y - G^H G p$ and $G^H G$, which is a stronger restriction than necessary.

Relative Doppler controls likelihood interactions

Theorem 1: common-offset invariance. Shift every physical Doppler by the same real δ, with the same delays, gains, prefix, and chirp parameters. Then

$$H_\delta = V_\delta H_0, \qquad V_\delta = A \Delta_\delta^+ A^H, \qquad V_\delta^H V_\delta = I, \qquad H_\delta^H H_\delta = H_0^H H_0. \tag{14}$$

Proof. C_l and Δ_δ are both diagonal, and $\Delta_{\nu+\delta} = \Delta_\delta \Delta_\nu$. Pull the common diagonal modulation to the left of the sum of path operators, then insert $A^H A = I$. Unitarity gives the final equality. No Gaussian approximation, integer δ, or truncation is involved.

Corollary 1. If all $\nu_r = k_r + \delta$ with integer k_r and integer $2Nc_1$, then R has at most $1 + P(P-1)$ cyclic interaction shifts per row. Self-products are diagonal; ordered cross-products supply at most $P(P-1)$ additional shifts and can themselves collide with the diagonal. Collisions can reduce the count. This is an upper bound, not a universal exact degree.

Proof. H_0 is a sum of P monomial operators. Each self-product is $|h_r|^2 I$. Each cross-product is another monomial operator, contributing at most one shift. Apply (14).

Corollary 2. For one path, $R = |h|^2 I$ even when H is dense. The channel likelihood then has no off-diagonal symbol interactions; only code constraints couple the symbols. For known CSI, full-model error probability is invariant to a common fractional offset because multiplying the observation by V_δ^H preserves proper white noise and returns the zero-common-offset channel.

These statements concern receiver interaction structure, not the difficulty of estimating unknown Dopplers. Unknown δ may still be difficult to infer from SP observations. Widely differing fractional offsets can make R dense. A large number of physical paths can also eliminate the apparent degree advantage, since the worst-case interaction count grows as P^2.

The explicit common-offset proof is robust to the CPP. More detailed attempts to rewrite each noninteger cross-product as a standard Doppler-delay atom must retain the circular wrap phases. This report does not silently discard those phases.

The most favorable common-offset experiment also has an exact classical solution: derotate by V_δ^H and detect using H_0. Accordingly, the theorem diagnoses an unnecessary representation cost; it does not establish a new receiver advantage over derotation.

Pointwise posterior and decision certificates

The prior chapter's joint-model KL identity assumes fixed generative channels. The present comparison instead freezes actual numerical scores after observing y. It therefore applies to a data-dependent plug-in $\bar{H}(y)$ without pretending that the estimate is exogenous. Its reference is the full **plug-in posterior**, not automatically the Bayesian posterior with channel uncertainty integrated out.

Let $\mathcal{C}$ be the allowed finite codebook, $P(d) = e^{s(d)}/Z$, $Q(d) = e^{s_B(d)}/Z_B$, and

$$e(d) = s(d) - s_B(d), \qquad \Omega = \max_{d\in\mathcal{C}} e(d) - \min_{d\in\mathcal{C}} e(d). \tag{15}$$

A constant score error has $\Omega = 0$ and no decision consequence.

Theorem 2: bounded score oscillation. For common positive codeword support,

$$\mathrm{TV}(P,Q) \le \tanh(\Omega/4), \qquad D_{\mathrm{KL}}(P\|Q) \le \Omega^2/8. \tag{16}$$

The total-variation constant is sharp over finite distributions.

Proof of the total-variation bound. Shift e by its minimum and let $W = e^e \in [1, K]$, $K = e^\Omega$. Under Q, put $m = \mathbb{E}_Q W$. Then $P/Q = W/m$ and $\mathrm{TV}(P,Q) = \mathbb{E}_Q|W - m|/(2m)$. By the chord bound for the convex function $|W - m|$ on $[1, K]$,

$$\mathrm{TV}(P,Q) \le \frac{(K-m)(m-1)}{m(K-1)}.$$

The right side is maximized at $m = \sqrt{K}$ and equals $(\sqrt{K}-1)/(\sqrt{K}+1) = \tanh(\Omega/4)$. A two-point distribution at the interval endpoints attains equality. The $K = 1$ case follows directly.

Proof of the KL bound. $Q = Pe^{-e}/\mathbb{E}_P e^{-e}$ implies $D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_P e + \log \mathbb{E}_P e^{-e}$. The bounded-variable exponential-moment inequality gives at most $\Omega^2/8$. For completeness, this inequality follows by bounding the centered variable's moment generating function by the two-endpoint Bernoulli interpolation

and integrating the bound that its log moment generating function has second derivative at most $\Omega^2/4$ from zero to one. These are general probability tools, not AFDM-specific inventions.

Corollary 3: bit-LLR fidelity. For an information bit taking both values in the codebook,

$$|L_P - L_Q| \le \Omega. \tag{17}$$

Proof. Within either bit subset, replacing s_B by $s_B + e$ shifts a log-sum-exp by a number between $\min e$ and $\max e$. Subtracting the two subset shifts gives a range of at most Ω.

Corollary 4: exact-MAP fidelity. Suppose $\widehat{d}_B$ is a global maximizer of s_B and its gap to the best other codeword is m_B. If

$$m_B > \Omega, \tag{18}$$

then $\widehat{d}_B$ is the unique full-score maximizer.

Proof. For every other d, $s(\widehat{d}_B) - s(d) \ge m_B - \Omega > 0$.

The global qualification is essential. A local BP belief margin, a syndrome check, or the difference between two codewords on an incomplete list is not a global m_B. A list-based application certifies only that list unless there is a valid bound on all omitted codewords. The new experiment uses complete enumeration precisely to avoid this hidden assumption.

A cheap conservative energy bound

If $E = R - R_B$ has zero diagonal and only selected Hermitian off-diagonal pairs, QPSK gives

$$\Omega \le \overline{\Omega}_B = \frac{4P_d}{\sigma^2} \sum_{i<j:\,(i,j)\ \text{omitted}} |R_{ij}|. \tag{19}$$

Proof. $d^H E d = 2\Re \sum_{i<j} d_i^* E_{ij} d_j$. Each product has magnitude $P_d|E_{ij}|$. The range across codewords is at most twice the absolute bound, giving (19).

If the linear statistic is also approximated with error Δz, a sufficient additional range penalty is $4\sqrt{P_d}\|\Delta z\|_1/\sigma^2$. This quantifies the cost of replacing Rp by $R_B p$ rather than merely saying that pilot cancellation matters.

At equal edge costs, retaining the largest $|R_{ij}|$ minimizes (19) for a fixed number of pairs. Thus the proposed conservative bound **reduces to classical magnitude pruning**, not a new ranking algorithm. A new high-dimensional support method would need a tighter code-aware bound or a different demonstrably useful objective.

Separating compression from channel-inference error

Let $P_\star(\cdot \mid y)$ denote the desired true message posterior, including channel uncertainty when appropriate, and $P_{\bar{H}}$ the full plug-in posterior. Triangle inequality yields

$$\mathrm{TV}(P_\star, Q_B) \le \mathrm{TV}(P_\star, P_{\bar{H}}) + \tanh(\Omega/4). \tag{20}$$

The first term is channel/inference error; the second is representation-induced score distortion. Compression certification only controls the second. If Q_B is globally decoded, its conditional decision regret relative to $P_\star$ is at most twice this TV bound, clipped at one. Approximate decoding adds another unbounded term unless separately analyzed. This decomposition prevents a correct compression theorem from being advertised as a complete solution of SP channel estimation.

Proposed reference method and guarantees

Certificate-guided likelihood sparsification

The method below is fully implemented for a finite codebook. It serves as a rigorous reference policy and a specification for what a future scalable approximation would have to preserve.

1. Obtain a channel estimate through a separately specified estimator; freeze it for this terminal comparison. Compute R and the exact z of (11).
2. Sort unordered off-diagonal pairs by decreasing magnitude. Keep the diagonal unchanged in the implementation; it could be omitted from QPSK scoring after computing z.
3. Start with B_0 retained pairs. Compute s_{B_0} on the complete codebook and $\overline{\Omega}_{B_0}$.
4. If the global score gap exceeds the bound, output the certified MAP decision. Otherwise add a predetermined batch of pairs and update scores.
5. Stop at the first certificate or use the complete model as a fallback. If a separate hard budget prevents fallback, the output must be labeled uncertified; the present executed policy uses full fallback.

The primary dense reference and a separately implemented sparse/incremental version produce identical decisions and stopping budgets on every saved oracle observation. The incremental version computes the linear scores once and evaluates each newly retained pair contribution once, rather than repeating dense products. Both preserve full-model MAP decisions because the stopping condition is sufficient and the fallback is exact.

This is not a proof that progressively adding edges monotonically improves realized FER. Only the conservative bound decreases monotonically. Nor does the policy improve on exact full-model MAP: it attempts to reproduce it with a smaller interaction representation.

Complexity and the unclosed computational condition

For an arbitrary dense estimated channel, forming R costs $O(N^3)$ and z costs $O(N^2)$. Fast physical-path matvecs can reduce applying H or H^H to $O(PN + N \log N)$, but that does not by itself establish cheap formation of every desired Gram entry. This package uses dense construction for small validation, not a production fast Gram constructor.

For a codebook of size M, the shared linear scores cost $O(MN)$ per observation. Evaluating B pairwise contributions costs $O(MB)$. Incremental addition avoids re-evaluating previous pairs, but finding the global margin repeatedly costs $O(M)$ per stage. Complete enumeration has $M = 2^k$ and is not scalable to production information length.

A graph with $N + 2B$ logical matrix entries is not necessarily cheaper than a channel represented by three integer atoms and $3N$ nonzero entries. Exact matched statistics, Gram preparation, edge ordering, decoder calls, and certificate acquisition must all be counted. The new comparison matches that simple entry-count proxy at $B = N$, but **does not claim equal wall-clock budgets**. Even the optimized implementation is slower than the full small-code kernel in the measured examples.

A deployable Transactions contribution would have to close both of these gaps: an affordable full-statistic path under estimated CSI, and a valid inexpensive decision/posterior fidelity assessment for the actual iterative decoder. Neither is supplied by relabeling the tiny-code enumerator as a low-complexity LDPC receiver.

Executed experiments

Design, controls, and raw evidence

The useful block lengths are 8 and 12; chirps use $c_1 = 3/(2N)$ and $c_2 = (\sqrt{2}-1)/(2N^2)$. The reused Gallager constructions have $(n, k) = (16, 6)$ and $(24, 8)$ coded/information bits. These are small members of the coding infrastructure, with substantial finite-size structure; they are not representative long LDPC codes. All MAP and posterior calculations enumerate all codewords.

Pilot fractions are $\{0.25, 0.5, 0.75\}$ and complex noise variances are $\{0.2, 0.5\}$. The master seed is 20260919; phase seeds are 11, 29, and 47. The pilot is a deterministic constant-envelope quadratic-phase vector. Path amplitude ratios are $1 : 0.65 : 0.40$, normalized once before propagation. All channel compression arms receive the same oracle channel and physical parameters; no method gets a different transmitted waveform or noise realization.

Regime	Delays	Normalized Dopplers
Single path	$[1]$	$[0.37]$
Common fractional offset	$[0, 1]$	$[0.37, 1.37]$
Nearby fractional offsets	$[0, 1]$	$[0.32, 1.42]$
Different offsets	$[0, 1]$	$[0.37, -0.29]$
Three paths	$[0, 1, 2]$	$[0.37, 1.12, -0.41]$

The controls are: full-channel codeword MAP; top-three integer channel coefficients; the same compressed data channel with the **full pilot response**; isotropic-$\mathcal{J}$ weighted projection; actual-code-second-moment weighted projection; magnitude-pruned Gram with $B = N$; the same Gram with compressed pilot linear term; common-offset derotation followed by top-three channel coefficients; conventional proper-isotropic LMMSE scalar likelihoods followed by exact codeword selection; and the certified reference.

These are strong analytic/oracle controls for the mechanism, not faithful implementations of every published receiver. No claim is made to have reproduced the numerical curves of MF-MP, D-GESBL, JSG, or HS-JCPS. Exact full-model MAP is the optimal message decision in the stated known-channel experiment, making downstream suboptimal detection an excluded explanation for the main comparison.

Structural result survives the collision control

At $N = 8$, the common-offset two-path Gram has 16 nonzero entries because the two cross shifts coincide at a half-block displacement. At $N = 12$, they do not coincide: the Gram has 36 entries. The full physical channel has 64 and 144 nonzero entries, respectively. The Gram remains exactly represented by the $B = N$ budget at both lengths. A single path has only N diagonal Gram entries. Nearby or unequal fractional offsets generally produce dense Grams.

Thus the common-offset finding is not an accidental $N = 8$ shift-collision artifact. The collision changes the degree, not the common-offset invariance theorem.

Coded results on the balanced grid

Each table entry below is the number of failed decoding instances out of **4,608 paired grid evaluations per regime**, aggregating three seeds, three pilot fractions, two noise variances, and 256 observations. This aggregation is descriptive. Reused observations across settings are not treated as independent trials; there are no binomial population confidence claims for these totals.

Block length $N = 8$.

Regime	Full MAP	H3-PC	Gram	Derotation	LMMSE
Single path	545	684	545	545	545
Common offset	564	1000	564	564	719
Nearby offsets	560	1034	565	581	733
Different offsets	575	874	673	699	767
Three paths	584	1023	690	825	807

Block length $N = 12$.

Regime	Full MAP	H3-PC	Gram	Derotation	LMMSE
Single path	403	597	403	403	403
Common offset	509	999	509	509	686
Nearby offsets	493	1024	509	522	691
Different offsets	500	901	617	693	691
Three paths	512	1009	694	785	728

Here H3-PC means three integer channel atoms with exact pilot cancellation; Gram means $B = N$ retained unordered interactions with exact z; derotation removes the first path's common fractional component before the same three-atom projection. The certified policy matches the full-model column exactly in all regimes, with variable final interaction budget.

The fixed Gram advantage over three channel atoms is large in several fixtures. It is nevertheless not a novel-method win: the common-offset result is matched exactly by classical derotation, and matched-filter AFDM detection is prior art [R5]. In the different-offset and three-path regimes, a fixed Gram budget loses decisions relative to the full model; no universal sparse-graph exactness is claimed.

SP linear-term and weighted-projection ablations

Preserving the pilot linear statistic matters. At $N = 12$, replacing Rp by the pruned R_{BP} increases the different-offset failure count from 617 to 890 and the three-path count from 694 to 1,137 on the same grid. This is not an extra channel-estimation benefit: both arms use the same known channel, and only the likelihood construction changes.

The first exploration used isotropic data covariance for the $\mathcal{J}$-weighted projection. The actual small codebooks do not have isotropic covariance. Rather than interpreting that surrogate as an exact coded objective, the skeptical pass added a second projection using the complete codebook second moment. The independent verifier confirms that this exact-code projection decreases the true code-averaged observation mismatch on every saved fixture. Nevertheless, its aggregate FER is worse than ordinary H3 in several regimes; for example, at $N = 12$, common-offset failures are 1,348 versus 1,318. In the

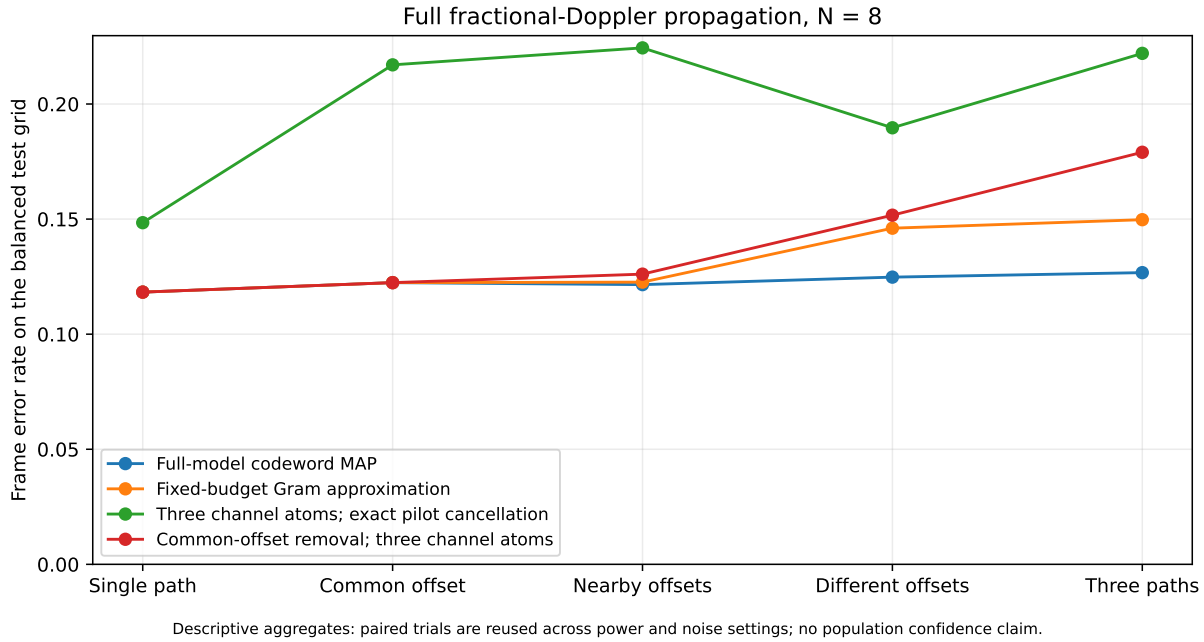


Figure 1: Balanced-grid coded comparison at N=8. These curves describe paired finite fixtures, not a channel-ensemble confidence statement.

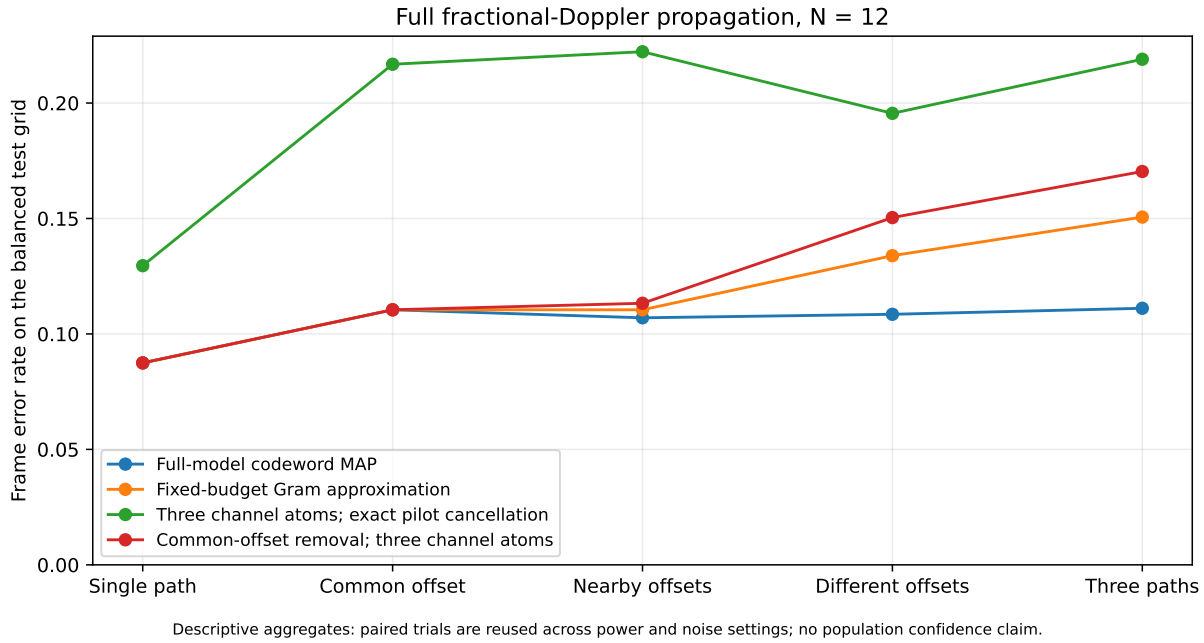


Figure 2: Second block-size control at N=12. The common-offset equality remains after the N=8 cross-shift collision is removed.

three-path regime it improves to 1,272 versus 1,325. It is therefore neither universally harmful nor an automatic communication-optimal representation.

The existence of a $\mathcal{J}$ decrease is not a proof of FER ordering. This conclusion survives correcting the covariance surrogate and is not a normalization or code-correlation bookkeeping artifact.

Certificate tightness and costs

N	Regime	Initially certified	Mean final pairs	Full-model fallback
8	Common offset	100.00%	8.00	0.00%
8	Nearby offsets	74.37%	10.36	1.32%
8	Different offsets	0.61%	19.72	11.00%
8	Three paths	1.19%	19.05	9.14%
12	Common offset	100.00%	12.00	0.00%
12	Nearby offsets	23.35%	26.46	0.30%
12	Different offsets	0.00%	45.44	1.19%
12	Three paths	0.00%	53.30	3.43%

The full number of unordered pairs is 28 at $N = 8$ and 66 at $N = 12$. At common offset, all decisions are certified immediately because the omitted energy is numerical zero. At $N = 12$ with nearby offsets, the universal bound initially certifies only 23.35% of observations; the exact codebook energy range certifies 82.07%. This demonstrates that bound conservatism is a major cost, not merely actual posterior sensitivity. Computing that exact range by full codeword enumeration is not free.

The first dense reference redundantly rescored previously retained interactions. The skeptical implementation pass removed that redundancy: sparse pair contributions and the linear term are now computed incrementally. All decisions and stopping budgets remain identical across the 540 equivalence checks. The improved implementation still incurs sorting, repeated global-margin evaluation, and preparation costs.

N	Regime	Full exact (microseconds)	Fixed sparse Gram	Incremental certificate
8	Common offset	13.28	64.76	66.37
8	Nearby offsets	13.32	65.14	65.68
8	Three paths	13.18	64.65	125.52
12	Common offset	33.22	81.05	82.21
12	Nearby offsets	32.91	82.51	116.93
12	Three paths	33.06	81.70	320.14

These are medians of 101 repeated **single-observation tiny-code kernels**, including preparation, on fixed representative inputs. They are not production timing confidence intervals. The result establishes only that a speedup has not been demonstrated by the implemented small-system policy. It does not

prove that a future sparse long-code implementation must be slower. Conversely, reduced final graph size is not evidence that such a speedup already exists.

Same-observation estimated-channel diagnostic

A separate known-delay, single-path estimator obtains its Doppler and complex gain by a 65-point pilot-only grid/profile-LS calculation. It uses the current observation but no transmitted data or true channel. The same resulting $\bar{H}(y)$ is supplied to the full plug-in decoder and its Gram representation.

At $N = 8$, the three seeds produce 71 errors in 384 frames with the full plug-in model, exactly 71 with the diagonal Gram, and 108 with three integer channel coefficients. The actual-channel oracle makes 2 errors on those same frames. At $N = 12$, the corresponding counts are 31, 31, 77, and 1. Every full-versus-Gram decision agrees, and centered scores agree within 2.85×10^{-13}.

This is a test of **pointwise plug-in equivalence under endogenous estimation**, not evidence that the pilot-only estimator competes with I1. The remaining channel-estimation gap is deliberately visible. Likelihood compression does not cure it.

Novelty audit and baseline boundary

The audit distinguishes a theorem being derived correctly in this report from that theorem being historically new. General bounded-energy posterior inequalities, the matched-filter likelihood, and magnitude pruning are classical mathematical ingredients. A keyword search that finds no identical title cannot establish novelty.

Closest work	Relevant overlap	Consequence for this project
Zheng et al., SP-AFDM [R1]	SP pilot placement and iterative CE/detection	SP and iterative cancellation are not contributions by themselves
Huang et al., SP-OTFS [R2]	GCE-BEM and low-cost joint message passing	Integer/BEM soft reception is established
Kanazawa et al. [R3]	Coded SP, soft decoder feedback, throughput/cost	I1-type feedback is a strong classical family
Luo et al., AFDM-ISAC SBL [R4]	SP, fractional parameters, data-aided SBL and lower-cost inference	Generic SP plus fractional/SBL/data feedback is occupied
Wu et al., MF-MP [R5]	Matched-filter matrix sparsity for fractional-Doppler AFDM	The central compression-domain switch is directly anticipated
Luo et al., JSG [R6]	Joint AFDM detection/LDPC graph and weak-edge simplification	Coded graph sparsification is not a new architecture
Mao et al., HS-JCPS [R7]	Chirp selection and structured sparsification under complexity constraints	A generic BER-aware sparse-receiver trade-off is already studied
Rusek and Prlja [R8]	Information-theoretic optimal channel shortening	Communication-aware representation is not new in general

The 2024 MF-MP paper is particularly decisive: its public abstract and introduction explicitly motivate matched filtering by the dense fractional-Doppler factor graph. Its full derivation was not accessible

through the publisher in this run, so no claim is made about whether every bound in this report is absent from its full text. Its stated contribution is already sufficient to reject "we first exploit a sparse matched-filter AFDM channel."

The July 22, 2026 HS-JCPS preprint adds a second relevant constraint: it explicitly co-designs structured sparsification and AFDM parameters under a computational limit, with a BER-related metric. The present manuscript must distinguish any future claim from that contribution rather than citing only older pilot-estimation papers.

The possible remaining contribution is narrower: a **usable, decoder-compatible fidelity criterion** that allocates interaction work while preserving the SP linear statistic, survives arbitrary relative Dopplers, and delivers a favorable total-cost curve against MF-MP, derotation, structured MMSE, and a frozen turbo receiver. The present score-oscillation theory specifies such a criterion at the exact finite-code level. Its scalable realization and publication novelty have not yet been demonstrated.

Skeptical review: what was challenged and what survived

The dominant alternative explanations were actually checked. The full physical model was compared with an independent sample/CPP construction; signs were mapped explicitly. A second block size removed a favorable shift collision. The code's nonisotropic covariance was detected and an exact-code weighted projection added. The full pilot response was held fixed to separate pilot cancellation from data-channel compression. Conventional derotation exactly reproduced the favorable common-offset gain. Exact codeword inference removed loopy detector/decoder quality as a confound. An observation-driven channel estimate established that the pointwise score guarantee does not require independent estimation data. A sparse/incremental implementation removed repeated dense score construction, while recording the remaining overhead.

The strongest surviving result is **structural and conditional**: interaction complexity depends on the relative channel geometry and the representation of the likelihood, not only on the number of integer atoms needed to approximate H. The strongest rejected interpretation is that this observation alone creates a new low-complexity SP-AFDM receiver. Direct prior art and a simpler exact control refute that interpretation.

Several boundaries remain. The tiny codes have unusual short-block dependencies. Only three channel-phase seeds and five controlled path configurations are in the discriminative grid. Proper white noise and ideal integer-delay sampled propagation are assumed. Full Gram acquisition is dense in the implementation. Exact global margins are exponentially expensive for general long codes. The adaptive guarantee is a hard-decision guarantee; it does not automatically certify the soft LLRs emitted by an approximate iterative decoder. Marginal LLR bounds are available only when the compared normalized scores are actually evaluated or otherwise accurately represented.

Remaining confirmation, not invented results

A complete executable independent-stream **small-code** confirmation driver is included. It uses new seeds, writes to a separate directory, and preserves the same radio/control definitions. It was not run because it would principally refine small-system statistical precision rather than change the main research verdict.

A separate production-confirmation configuration records $N = 1024$, full G1, frozen I1 and same-support ridge controls, decoder-call equality, complete setup/application timing, failure diagnostics, and required source/checkpoint fingerprints. It is explicitly **not executed**. The partial handoff cannot run that

configuration end to end without the missing project modules/checkpoint and an audited terminal adapter. No dummy implementation is substituted for them.

A future production method must solve a computational issue, not simply gather more favorable toy counts. If an approximate BP or list decoder supplies its own margin, that quantity must be related to the full code's score with a valid omitted-candidate bound, or the guarantee must be relabeled as heuristic. Likewise, retaining an exact z must be paid for; a method that silently acquires a richer channel to form it has a different information budget.

Claim-to-evidence map and final judgment

Claim	Status	Main evidence
C01: G1 and manuscript convention reconcile	Proved + numerical	Eq. (4); sign convention section
C02: Common offset cancels from Gram	Proved + numerical	Theorem 1
C03: Sparse likelihood can preserve dense-channel decoding	Proved special case + numerical	Proposition 3 and Corollary 1
C04: Score oscillation bounds posterior/LLR/MAP	Proved + numerical	Theorem 2 and Corollaries 3-4
C05: Adaptive policy preserves full-model MAP	Proved with full fallback + numerical	Eq. (18); algorithm section
C06: Data-dependent estimate permits pointwise score equality	Proved + numerical	Eq. (12),(20)
C07: TAIL has no established advantage over same-support ridge	Source reported	Source audit
C08: Matched-filter AFDM sparsity is prior art	Literature supported	Novelty audit [R5]
C09: No measured small-code cost advantage	Empirical scope only	Optimized timing table
C10: Smaller coded observation mismatch need not lower empirical FER	Numerical counterexample	Weighted-projection ablation
C11: Diagonal Gamma collapses, correlations can matter	Proved + numerical	Proposition 2
C12: New production Transactions advantage	Unconfirmed	Research verdict

The report, source code, configurations, raw fixtures, results, negative controls, timing records, and hashes are delivered together. The earlier 26-page chapter should be retained as a foundation, with its G1 evidence-status statements updated from the new handoff. Its many correct identities should not be converted into a contribution list without a defensible comparison against prior art.

Final judgment. Select likelihood fidelity under constrained representation as the single research line, but do not submit the certified reference or the old tail candidate as a proven new Transactions receiver. The mathematical mechanism is now sharper, the full-propagation controlled evidence is executable, the strongest obvious classical explanation has been tested, and the current bottleneck is explicit: producing

useful, affordable fidelity control for the actual coded iterative receiver. This is the narrow unresolved method question that remains after the weaker narratives have been removed.

References and source record

Primary public sources were checked on September 19, 2026. Publication metadata and access levels are stated rather than silently upgrading a preprint to a journal result.

[**R1**] K. Zheng, M. Wen, T. Mao, L. Xiao, and Z. Wang, *Channel Estimation for AFDM With Superimposed Pilots*, arXiv:2404.10232, April 16, 2024. Abstract and manuscript record checked; original chapter contains its detailed pilot-model treatment.

[**R2**] F. Huang, Q. Guo, Y. Zhang, and Y. Zakharov, *Message Passing-Based Joint Channel Estimation and Signal Detection for OTFS with Superimposed Pilots*, arXiv:2309.08177v2, April 15, 2024; IEEE Transactions on Vehicular Technology, DOI 10.1109/TVT.2024.3378833. HTML and record consulted.

[**R3**] Y. Kanazawa, H. Iimori, C. Pradhan, S. Malomsoky, and N. Ishikawa, *Superimposed Pilot-Based OTFS: Will It Work?*, arXiv:2501.15935v3, February 24, 2026; IEEE Transactions on Vehicular Technology, 2026, DOI 10.1109/TVT.2025.3608199. Current record and soft-feedback scope checked.

[**R4**] Y. Luo et al., *Data-aided Channel Estimation and Sensing With Sparse Bayesian Learning for AFDM-ISAC System*, arXiv:2607.18881v1, July 21, 2026. Full HTML consulted. Used as close SP/off-grid/data-aided prior art; its curves were not reproduced.

[**R5**] L. Wu, S. Luo, D. Song, F. Yang, R. Lin, and S. Xie, *AFDM signal detection based on message passing scheme*, Digital Signal Processing, vol. 153, article 104633, October 2024. DOI 10.1016/j.dsp.2024.104633. Publisher-indexed abstract, introduction, authorship, and metadata checked; full publisher fetch returned 403.

[**R6**] Q. Luo et al., *Joint Sparse Graph for Enhanced MIMO-AFDM Receiver Design*, arXiv:2503.19143v1, March 24, 2025. HTML and current record consulted.

[**R7**] R. Mao et al., *Joint Chirp Parameter Selection and Low-Complexity MMSE Receiver Design for AFDM Systems*, arXiv:2607.20227v1, July 22, 2026. Full HTML consulted; used as structured-sparsification/complexity prior art.

[**R8**] F. Rusek and A. Prlja, *Optimal Channel Shortening for MIMO and ISI Channels*, IEEE Transactions on Wireless Communications, vol. 11, no. 2, pp. 810–818, 2012. DOI 10.1109/TWC.2011.121911.110809. Author-institution publication record and abstract consulted.

[**P1**] Supplied *A Consistent Theory of Coded SP-AFDM Reception*, September 19, 2026, 26 pages.

[**P2**] Handoff `MANIFEST.txt`, `STATE.md`, `DERIVATION_PACKAGE.md`, `research-protocol.md`, and `l16-reassessment-v1.md`. Exact copies and hashes are included under `inputs/`.

[**P3**] Handoff `tail-terminal-results-v5.md` and `tail-recheck-results-20260917.md`. Numerical production claims are attributed to these reports, not to the new simulation.